

Problem 1

Given the following twist ξ , what is the homogeneous transformation matrix?

(a) $\xi = [v^\top \ \Omega^\top]^\top = [3/\pi \ 0 \ 0 \ \pi/3 \ 0 \ 0]^\top.$

(b) $\xi = [v^\top \ \Omega^\top]^\top = [0 \ 0.5 \ 0 \ 0 \ \pi/2 \ 0]^\top.$

(c) $\xi = [v^\top \ \Omega^\top]^\top = [1 \ 0 \ 0 \ 0 \ 0 \ \pi/3]^\top.$

Part (a)

$$T = \begin{bmatrix} 1.0000 & 0 & 0 & 1.0000 \\ 0 & 0.5000 & -0.8660 & 0 \\ 0 & 0.8660 & 0.5000 & 0 \\ 0 & 0 & 0 & 1.0000 \end{bmatrix}$$

Part (b)

$$T = \begin{bmatrix} 0 & 0 & 1.0000 & 0 \\ 0 & 1.0000 & 0 & 0.7854 \\ -1.0000 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1.0000 \end{bmatrix}$$

Part (c)

$$T = \begin{bmatrix} 0.5000 & -0.8660 & 0 & 0.8660 \\ 0.8660 & 0.5000 & 0 & 0.5000 \\ 0 & 0 & 1.0000 & 0 \\ 0 & 0 & 0 & 1.0000 \end{bmatrix}$$

Problem 2

Given the homogeneous transformation matrix,

$$\mathbf{T} = \begin{bmatrix} 0.5000 & -0.6124 & 0.6124 & 0.8415 \\ 0.6124 & 0.7500 & 0.2500 & 0.3251 \\ -0.6124 & 0.2500 & 0.7500 & -0.3251 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- (a) What is the displacement?
- (b) What is the Quaternion that encodes the rotation?
- (c) What is the Angle-Axis that encodes the rotation?
- (d) What is the twist ξ that encodes the full transform?

Part (a): Displacement

The displacement is the upper-right 3×1 block of $\mathbf{T}$:

$$\mathbf{d} = \begin{bmatrix} 0.8415 \\ 0.3251 \\ -0.3251 \end{bmatrix}$$

Part (b): Quaternion

With $\text{tr}(\mathbf{R}) = 0.5 + 0.75 + 0.75 = 2$:

$$\begin{aligned} q_0 &= \sqrt{\frac{1 + \text{tr}(\mathbf{R})}{4}} = \sqrt{\frac{3}{4}} = 0.866 \\ q_1 &= \frac{r_{32} - r_{23}}{4q_0} = \frac{0.25 - 0.25}{4(0.866)} = 0 \\ q_2 &= \frac{r_{13} - r_{31}}{4q_0} = \frac{0.6124 - (-0.6124)}{4(0.866)} = 0.354 \\ q_3 &= \frac{r_{21} - r_{12}}{4q_0} = \frac{0.6124 - (-0.6124)}{4(0.866)} = 0.354 \end{aligned}$$

Check: $q_0^2 + q_1^2 + q_2^2 + q_3^2 = 0.75 + 0 + 0.125 + 0.125 = 1$.

$$\mathbf{Q} = 0.866 + \begin{bmatrix} 0 \\ 0.354 \\ 0.354 \end{bmatrix}$$

Part (c): Angle-Axis

The vector of off-diagonal differences is

$$\begin{bmatrix} r_{32} - r_{23} \\ r_{13} - r_{31} \\ r_{21} - r_{12} \end{bmatrix} = \begin{bmatrix} 0 \\ 1.2248 \\ 1.2248 \end{bmatrix}, \quad \|\cdot\|_2 = \sqrt{0^2 + 1.2248^2 + 1.2248^2} = 1.7321.$$

Then

$$\theta = \text{atan2}\left(\frac{1}{2} \|\cdot\|_2, \frac{\text{tr}(\mathbf{R}) - 1}{2}\right) = \text{atan2}(0.866, 0.5) = \frac{\pi}{3} \text{ rad}$$

$$\hat{\mathbf{k}} = \frac{1}{2 \sin \theta} \begin{bmatrix} 0 \\ 1.2248 \\ 1.2248 \end{bmatrix} = \frac{1}{1.7321} \begin{bmatrix} 0 \\ 1.2248 \\ 1.2248 \end{bmatrix} = \begin{bmatrix} 0 \\ 0.707 \\ 0.707 \end{bmatrix}$$

$$\mathbf{\Omega} = \theta \hat{\mathbf{k}} = \frac{\pi}{3} \begin{bmatrix} 0 \\ 0.707 \\ 0.707 \end{bmatrix} = \begin{bmatrix} 0 \\ 0.741 \\ 0.741 \end{bmatrix}$$

Part (d): Twist

The angular part is $\mathbf{\Omega}$ from Part (c). The linear part follows the course's inverse-twist formula (with $\boldsymbol{\xi} = [\mathbf{v}; \theta \hat{\mathbf{k}}]$, where $\mathbf{v}$ is the linear part per radian):

$$\mathbf{v} = \left(\frac{\sin \theta}{2(1 - \cos \theta)} \mathbf{I} + \frac{2(1 - \cos \theta) - \theta \sin \theta}{2\theta(1 - \cos \theta)} \hat{\mathbf{k}} \hat{\mathbf{k}}^\top - \frac{1}{2} [\hat{\mathbf{k}}]_\times \right) \mathbf{d}.$$

For $\theta = \pi/3$: $\frac{\sin \theta}{2(1 - \cos \theta)} = 0.866$, so

$$\mathbf{v} = \left(0.866 \mathbf{I} + 0.0889 \hat{\mathbf{k}} \hat{\mathbf{k}}^\top - \frac{1}{2} \begin{bmatrix} 0 & -0.707 & 0.707 & 0.707 & 0 & 0 & -0.707 & 0 & 0 \end{bmatrix} \right) \begin{bmatrix} 0.8415 & 0.3251 & -0.3251 \end{bmatrix}.$$

Since $\hat{\mathbf{k}}^\top \mathbf{d} = 0.707(0.3251) + 0.707(-0.3251) = 0$, the $\hat{\mathbf{k}} \hat{\mathbf{k}}^\top$ term drops out, and

$$\mathbf{v} = 0.866 \begin{bmatrix} 0.8415 & 0.3251 & -0.3251 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} -0.4597 & 0.5950 & -0.5950 \end{bmatrix} = \begin{bmatrix} 0.9586 & -0.016 & 0.016 \end{bmatrix}.$$

$$\boldsymbol{\xi} = [\mathbf{v} \ \mathbf{\Omega}] = \begin{bmatrix} 0.959 & -0.016 & 0.016 & 0 & 0.741 & 0.741 \end{bmatrix}^\top$$

Problem 3

For the following give one pro and one con for using it as a representation,

- (a) Rotation and displacement separately.
- (b) Homogeneous transformation matrices.
- (c) Twist coordinates.

Part (a): Rotation and Displacement Separately

Pro: Easy to identify what is happening.

Con: Rotation and displacement are separate objects.

Part (b): Homogeneous Transformation Matrices

Pro: : composition is a single matrix multiply, so chaining frames is easy.

Con: Does not invert cleanly.

Part (c): Twist Coordinates

Pro: Provides velocity of object.

Con: It is singular or ambiguous at $\theta = 0$ and $\theta = \pi$

Problem 4

Given the following two homogeneous transformations, use MATLAB to plot a linear interpolation between the two. Your plot should contain a minimum of 5 intermediate points and should have the orthogonal axis (columns of the rotation matrix) drawn at each point. *Hint*: Use the `quiver3` function.

$$\mathbf{T}_{\text{initial}} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{T}_{\text{final}} = \begin{bmatrix} 0.5000 & -0.6124 & 0.6124 & 0.8415 \\ 0.6124 & 0.7500 & 0.2500 & 0.3251 \\ -0.6124 & 0.2500 & 0.7500 & -0.3251 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- (a) Perform the interpolation using ZYZ angles and position.
- (b) Perform the interpolation using Angle-Axis and position.
- (c) Perform the interpolation using Quaternion and position (remember to enforce the length constraint on the quaternion before calculating the intermediate orientations).
- (d) Perform the interpolation using twist.
- (e) Comment on the similarities or differences.

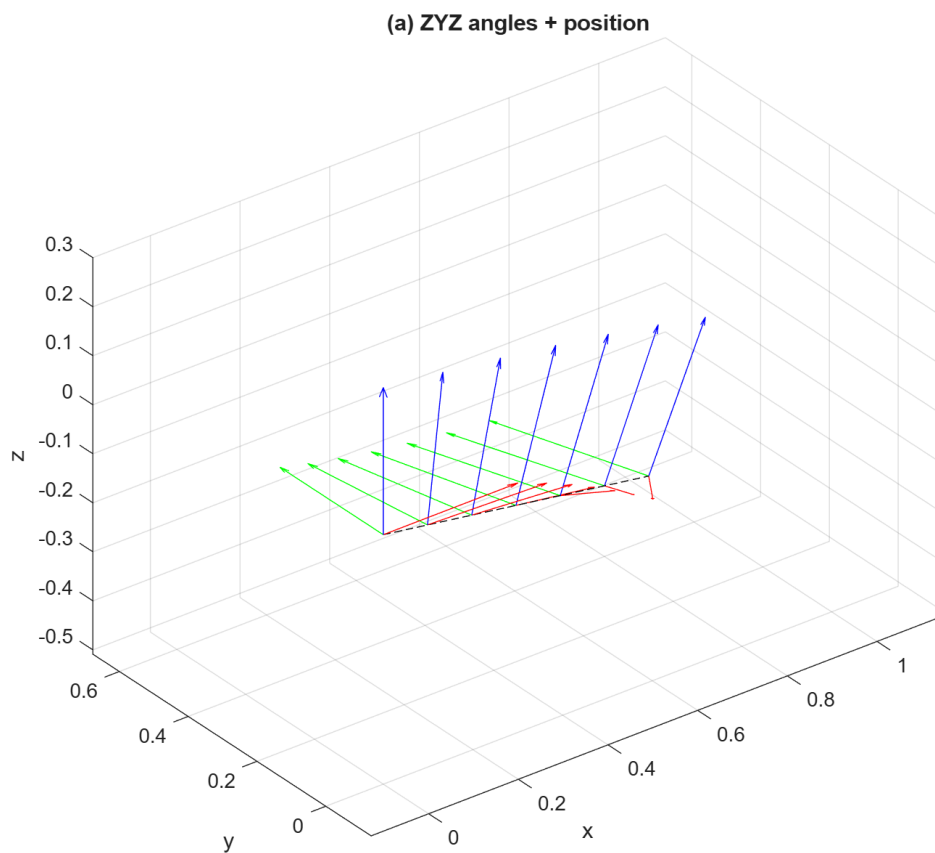
Part (a): ZYZ Interpolation

Figure 1: ZYZ-angle and linear-position interpolation.

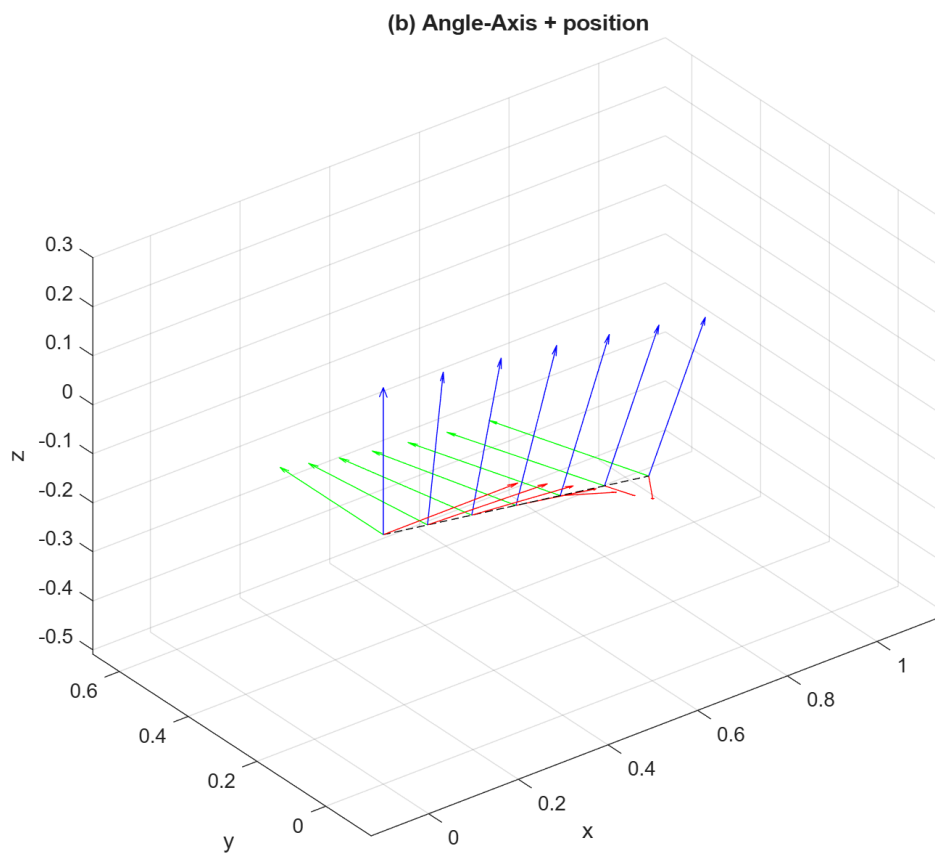
Part (b): Angle-Axis Interpolation

Figure 2: Angle-axis and linear-position interpolation.

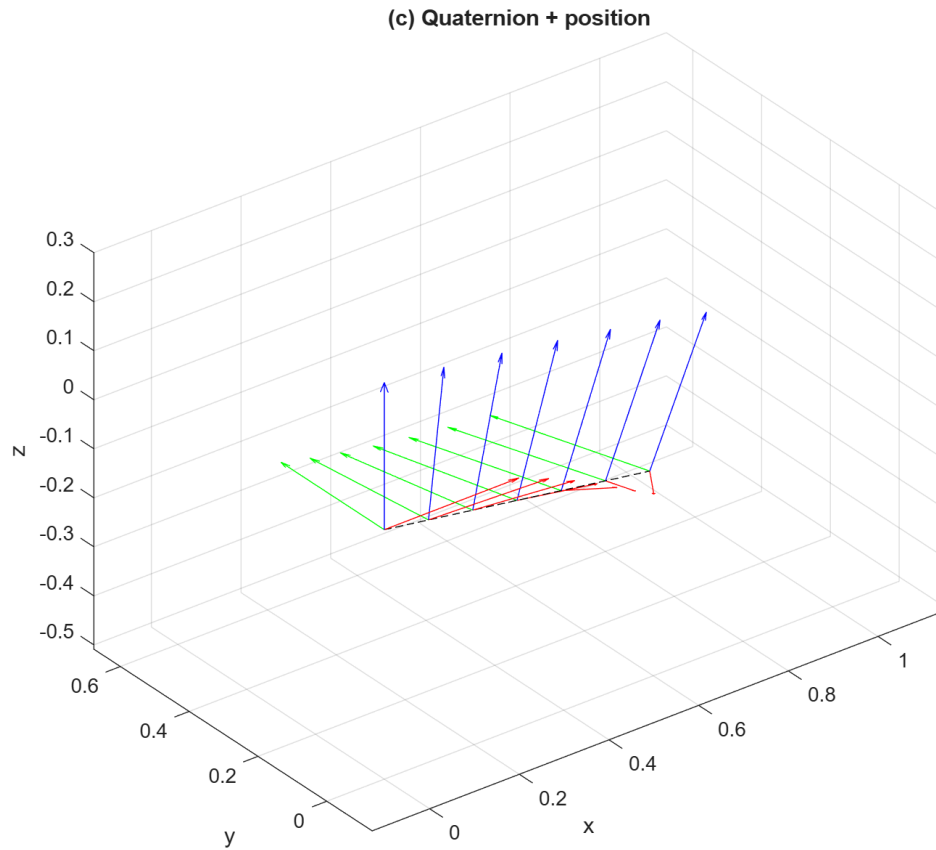
Part (c): Quaternion Interpolation

Figure 3: Normalized-quaternion and linear-position interpolation.

Part (d): Twist Interpolation

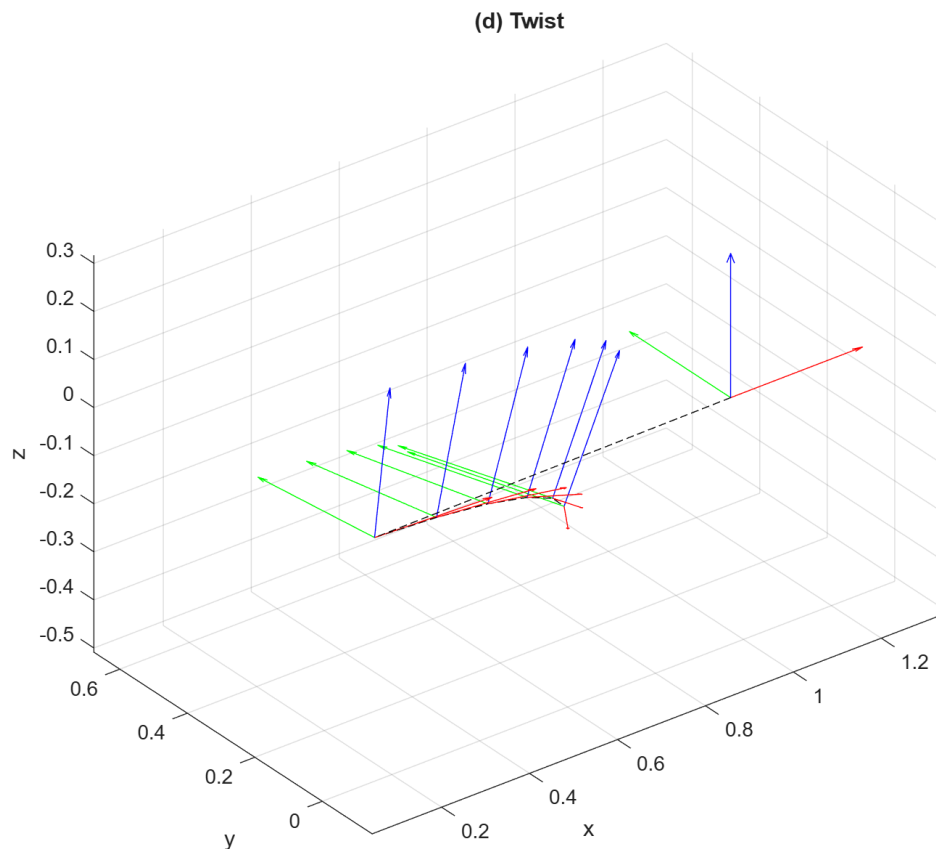


Figure 4: Twist interpolation.

Part (e): Comparison

All four start at the identity and end at $\mathbf{T}_{\text{final}}$, so the only real difference is what happens in between.

Angle-axis, quaternion, and twist all look basically the same for orientation. That makes sense since they all rotate about the same fixed axis $\hat{\mathbf{k}} = [0, 0.707, 0.707]^T$ by a total of $\pi/3$. The quaternion one is just a linear interpolation that gets renormalized, so its speed isn't perfectly constant, but for a 60° rotation you can't really tell.

ZYZ is the odd one out for orientation. It interpolates the three angles separately, so the frame doesn't spin about one axis and the intermediate orientations look different from the other three. It's also not the shortest path, and it has the singularity issue when the middle angle is 0 or π .

For position, twist is the odd one out. Angle-axis, quaternion, and ZYZ all use linear position, so the origin goes in a straight line. The twist does the rotation and translation together as a screw motion, so the origin curves along a helix instead.

Overall: angle-axis, quaternion, and twist agree on orientation, ZYZ doesn't, and twist is the only one that doesn't move in a straight line.

Appendix A: Code

Problem 1

Listing 1: Problem1.m

```
1 root = fileparts(fileparts(fileparts(fileparts(mfilename('fullpath')))));
2 addpath(fullfile(root, 'Functions'));
3
4
5 % Twists are  $\xi = [v; \omega]$ 
6 xi_a = [3/pi; 0; 0; 0; pi/3; 0; 0];
7 xi_b = [0; 0.5; 0; 0; 0; pi/2; 0];
8 xi_c = [1; 0; 0; 0; 0; 0; pi/3];
9
10 T_a = twist2Transform(xi_a);
11 T_b = twist2Transform(xi_b);
12 T_c = twist2Transform(xi_c);
13
14 disp('(a) T ='); disp(T_a);
15 disp('(b) T ='); disp(T_b);
16 disp('(c) T ='); disp(T_c);
```

Problem 2

[Problem 2.py not yet written]

Problem 3

[Problem 3.py not yet written]

Problem 4

Listing 2: Problem4.m

```

1 root = fileparts(fileparts(fileparts(fileparts(mfilename('fullpath')))));
2 addpath(fullfile(root, 'Functions'));
3 figDir = fullfile(fileparts(mfilename('fullpath')), '..', 'Latex', '
    Figures');
4 if ~exist(figDir, 'dir'), mkdir(figDir); end
5
6 T_i = eye(4);
7 T_f = [ 0.5000 -0.6124  0.6124  0.8415;
8         0.6124  0.7500  0.2500  0.3251;
9        -0.6124  0.2500  0.7500 -0.3251;
10         0        0        0        1    ];
11
12 R_i = T_i(1:3,1:3);  p_i = T_i(1:3,4);
13 R_f = T_f(1:3,1:3);  p_f = T_f(1:3,4);
14
15 N = 5;                                     % intermediate points (minimum 5)
16 s = linspace(0, 1, N + 2);               % includes both endpoints
17
18 %% (a) ZYZ angles + position
19 [phi, theta, psi] = rot2ZYZ(R_i);  zyz_i = [phi(1); theta(1); psi(1)];
20 [phi, theta, psi] = rot2ZYZ(R_f);  zyz_f = [phi(1); theta(1); psi(1)];
21 Ra = cell(1, numel(s));  pa = zeros(3, numel(s));
22 for k = 1:numel(s)
23     zyz = zyz_i + s(k) * (zyz_f - zyz_i);
24     Ra{k} = rotZ(zyz(1)) * rotY(zyz(2)) * rotZ(zyz(3));
25     pa(:,k) = p_i + s(k) * (p_f - p_i);
26 end
27 plotFrames(Ra, pa, '(a) ZYZ angles + position', fullfile(figDir, 'p4a.png'
    ));
28
29 %% (b) Angle-axis + position
30 om_i = rot2AngleAxis(R_i);
31 om_f = rot2AngleAxis(R_f);
32
33 Rb = cell(1, numel(s));  pb = zeros(3, numel(s));
34 for k = 1:numel(s)
35     om = om_i + s(k) * (om_f - om_i);
36     Rb{k} = angleAxis2Rot(om); % TODO
37     pb(:,k) = p_i + s(k) * (p_f - p_i);
38 end
39 plotFrames(Rb, pb, '(b) Angle-Axis + position', fullfile(figDir, 'p4b.png'
    ));
40
41 %% (c) Quaternion + position
42 % TODO: interpolate quaternions (rot2Quat / quat2Rot), enforce unit length
43 % BEFORE converting each intermediate quaternion to a rotation
44 Rc = cell(1, numel(s));  pc = zeros(3, numel(s));
45 Q_i = rot2Quat(R_i);
46 Q_f = rot2Quat(R_f);
47

```

```

48 if dot(Q_i , Q_f) < 0
49     Q_f = -Q_f;
50 end
51
52 for k = 1:numel(s)
53     Q = Q_i + s(k) * (Q_f-Q_i);
54     Q = Q / norm(Q);
55     Rc{k} = quat2Rot(Q);
56     pc(:,k) = p_i + s(k) * (p_f - p_i);
57 end
58 plotFrames(Rc, pc, '(c) Quaternion + position', fullfile(figDir, 'p4c.png'
59 ));
60 %% (d) Twist
61 % TODO:
62 x_i = transform2Twist(T_f);
63 Rd = cell(1, numel(s)); pd = zeros(3, numel(s));
64 for k = 1:numel(s)
65     x = [x_i(1:3) ; s(k)* x_i(4:6)];
66     T = twist2Transform(x);
67     Rd{k} = T(1:3,1:3);
68     pd(:,k) = T(1:3, 4);
69 end
70 plotFrames(Rd, pd, '(d) Twist', fullfile(figDir, 'p4d.png'));
71
72 %% (e) Comments
73 % TODO: comment on similarities / differences
74
75 function plotFrames(Rs, ps, ttl, file)
76     figure; hold on; grid on; axis equal;
77     plot3(ps(1,:), ps(2,:), ps(3,:), 'k--');
78     for k = 1:numel(Rs)
79         R = Rs{k}; p = ps(:,k);
80         quiver3(p(1), p(2), p(3), R(1,1), R(2,1), R(3,1), 0.3, 'r');
81         quiver3(p(1), p(2), p(3), R(1,2), R(2,2), R(3,2), 0.3, 'g');
82         quiver3(p(1), p(2), p(3), R(1,3), R(2,3), R(3,3), 0.3, 'b');
83     end
84     xlabel('x'); ylabel('y'); zlabel('z'); title(ttl);
85     view(3); hold off;
86     exportgraphics(gcf, file, 'Resolution', 200);
87 end

```

Appendix B: Hand Calculations

Problem 2

$$T = \begin{bmatrix} .5 & -.6124 & .6124 & .8415 \\ .6124 & .75 & .25 & .3251 \\ -.6124 & .25 & .75 & -.3251 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

a) $\underline{d} = \begin{bmatrix} .8415 \\ .3251 \\ -.3251 \end{bmatrix}$

b) $a_0 = \pm \sqrt{\frac{1 + \overset{2}{\text{Tr}(R)}}{4}} = .866$

$$a_1 = \frac{r_{32} - r_{23}}{4a_0} = \frac{.25 - .25}{4 \cdot .866} = 0$$

$$a_2 = \frac{r_{13} - r_{31}}{4a_0} = \frac{.6124 - (-.6124)}{4 \cdot .866} = .354$$

$$a_3 = \frac{r_{21} - r_{12}}{4a_0} = \frac{.6124 - (-.6124)}{4 \cdot .866} = .354$$

$$Q = .866 + \begin{bmatrix} 0 \\ .354 \\ .354 \end{bmatrix}$$

$$C) \quad \theta = \arctan 2 \left(\frac{1}{2} \left\| \begin{bmatrix} r_{32} - r_{23} \\ r_{13} - r_{31} \\ r_{21} - r_{12} \end{bmatrix} \right\|_2, \frac{\pm r(R) - 1}{2} \right)$$

$$\begin{array}{l} 0 \\ 1.2248 \leftarrow .25 - .25 \\ 1.2248 \quad .6124 - (-.6124) \\ \quad .6124 - (-.6124) \end{array}$$

$$\sqrt{0^2 + 1.225^2 + 1.225^2} = 1.73241$$

$$\theta = \arctan 2(.866, .5)$$

$$\theta = \pi/3 \text{ rad}$$

$$\hat{K} = \frac{1}{2.5\theta} \begin{bmatrix} r_{32} - r_{23} \\ r_{13} - r_{31} \\ r_{21} - r_{12} \end{bmatrix} = \frac{1}{2.5\theta} \begin{bmatrix} 0 \\ 1.225 \\ 1.225 \end{bmatrix}$$

$$\hat{K} = \begin{bmatrix} 0 \\ .637 \\ .637 \end{bmatrix}$$

$$\underline{\underline{\hat{K}}} = \frac{\pi}{3} \cdot \begin{bmatrix} 0 \\ .707 \\ .707 \end{bmatrix}$$

$$d) \underline{v} = \left(\frac{s\theta}{2(1-\cos\theta)} \underline{I} + \frac{2(1-\cos\theta) - \theta s\theta}{2\theta(1-\cos\theta)} \hat{k} \hat{k}^T - \frac{1}{2} [\hat{k}]_{\times} \right) \underline{d}$$

$$= \left(\frac{s \frac{\pi}{3}}{2(1-\cos \frac{\pi}{3})} \underline{I} + \frac{2(1-\cos \frac{\pi}{3}) - \frac{\pi}{3} s \frac{\pi}{3}}{2 \frac{\pi}{3} (1-\cos \frac{\pi}{3})} \begin{bmatrix} 0 \\ .707 \\ .707 \end{bmatrix} \begin{bmatrix} 0 \\ .707 \\ .707 \end{bmatrix}^T - \frac{1}{2} \begin{bmatrix} 0 & -.707 & .707 \\ .707 & 0 & 0 \\ -.707 & 0 & 0 \end{bmatrix} \right) \cdot \begin{bmatrix} .8415 \\ .3251 \\ -.3251 \end{bmatrix}$$

$$\underline{v} = [.9586, -.016, .016]^T$$

$$\underline{\omega} = \underline{\Omega} = \theta \cdot \hat{k} = \begin{bmatrix} 0 \\ .74 \\ .74 \end{bmatrix}$$

$$\underline{\xi} = \begin{bmatrix} .959 \\ -.016 \\ .016 \\ 0 \\ .74 \\ .74 \end{bmatrix}$$

Remainder of problems were
coded / written in latex